

System Identification of a SISO and MIMO System with Saturation and Butterworth Pre-filter

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Abstract—This paper presents the identification of a linear time-invariant system from input-output data, following the system identification cycle. We first characterize the saturation nonlinearity and Butterworth pre-filter in the signal path. Next, we perform nonparametric identification using spectral analysis of periodic multisine data to obtain a frequency response function (FRF) estimate. We then identify a parametric Box-Jenkins (BJ) model that provides a consistent estimate of the plant transfer function, and verify the asymptotic variance estimates through Monte Carlo simulations. Finally, we extend the approach to a 2×2 multiple-input multiple-output (MIMO) system using a Kung (Ho–Kalman) realization from noisy Markov parameter estimates. Throughout, we motivate the choices for input signals, model structures, and validation methods.

I. INTRODUCTION

ACCURATE mathematical models of dynamical systems are essential for control design, monitoring, and prediction. When first-principles models are unavailable or insufficiently accurate, system identification provides a data-driven alternative by constructing models directly from measured input-output data [1]. The identification process follows a systematic cycle: experiment design, data collection, model structure selection, parameter estimation, and model validation [1, 2]. Each step involves choices that affect the quality of the final model, and careful motivation of these choices is a central theme of this work.

Section II characterizes the saturation and Butterworth filter. Section III presents the nonparametric identification. Section IV covers parametric identification and validation. Section V compares simulated and theoretical variance. Section VI treats the MIMO case. We draw conclusions in Section VII.

II. SATURATION AND BUTTERWORTH FILTER

In this section, we characterize the known elements in the signal path from reference $r(k)$ to plant input $u(k)$ and discuss their impact on the identification of the unknown plant $G_0(q)$.

A. System Setup

The output $y(k)$ of the system is

$$y(k) = G_0(q)u(k) + H_0(q)e(k), \quad (1)$$

where $G_0(q)$ is the unknown plant transfer function, $H_0(q)$ is a stable monic noise filter, $e(k)$ is zero-mean white noise, $u(k)$ is the plant input, and q is the forward shift operator. The signal path from the user-prescribed reference $r(k)$ to $u(k)$ consists of a Butterworth low-pass filter $F(q)$ followed by a hard saturation $S(\cdot)$, as shown in Fig. 1.

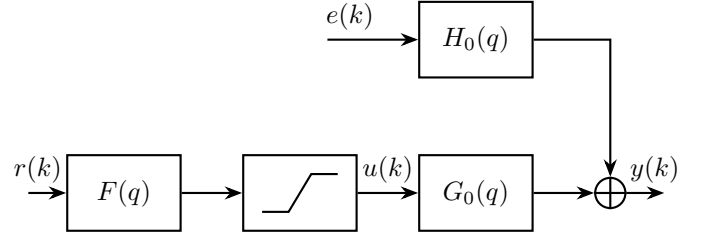


Fig. 1. Block diagram of the SISO system. The reference $r(k)$ passes through the Butterworth filter $F(q)$ and the saturation $S(\cdot)$ before entering the plant $G_0(q)$.

The Butterworth filter is given by

$$F(q) = \frac{0.7157 + 1.4315q^{-1} + 0.7157q^{-2}}{1 + 1.3490q^{-1} + 0.5140q^{-2}}, \quad (2)$$

a second-order filter that crosses -3 dB at 2.66 rad s^{-1} . Above the bandwidth the magnitude decays with 40 dB per decade. The saturation function is defined as

$$S(x) = \begin{cases} M & \text{if } x \geq M, \\ x & \text{if } -M < x < M, \\ -M & \text{if } x \leq -M, \end{cases} \quad (3)$$

where $M > 0$ is an unknown parameter that we must determine experimentally.

B. Determining the Saturation Parameter

To find M , we apply a slow linear sweep $r(k) \in [-5, 5]$ over $N = 100$ samples. Since this signal varies slowly and therefore has low frequency content, $F(q)$ passes it nearly unchanged, and $u(k)$ clips at $\pm M$ once the input to $S(\cdot)$ exceeds the saturation bound. We then extract

$$M = \max_k \{u(k)\}. \quad (4)$$

C. Influence on Identification

That the input is passed through a Butterworth low-pass filter and a saturation, has some implications on the identification of the plant G_0 :

- 1) **Bandwidth:** $F(q)$ is a second-order low-pass filter that attenuates high-frequency content in $r(k)$. This reduces the excitation energy, and hence also the signal-to-noise ratio (SNR) above its cutoff frequency of 2.66 rad s^{-1} . This may result in less accurate identification of the high-frequency system dynamics. The bandwidth lies close to

the Nyquist frequency of 3.14 rad s^{-1} , so the attenuated frequency range is limited.

- 2) **Amplitude:** The saturation limits $|u(k)| \leq M$, bounds the maximum power that can be injected into the system. All reference signals must therefore be designed with amplitude $\leq M$ at the output of $F(q)$ to avoid clipping.
- 3) **Distortion:** If $S(\cdot)$ activates, $u(k)$ becomes a nonlinear function of $r(k)$, strongly distorting the spectrum of the input away from the desired reference. We avoid this by scaling the reference amplitude to stay within $[-M, M]$.
- 4) **Identification target:** Since we identify G_0 from $u(k)$ to $y(k)$ (i.e., after the saturation), neither $F(q)$ nor $S(\cdot)$ are part of G_0 and do not need to be modeled.

III. NONPARAMETRIC IDENTIFICATION

In this section, we design an experiment and use nonparametric identification to obtain an FRF estimate of $G_0(q)$ and a noise spectrum. Information about the plant and noise behavior is inferred from these to inform the model structure selection in subsequent parametric identification.

A. Experiment Design

For nonparametric identification, we design an experiment of length $N = 1500$ samples, by combining $n = 10$ periods of $p = 150$ samples each. As the reference signal, we use a periodic Schroeder phase multisine input. The multisine is given by

$$r(k) = M \sum_{l=1}^L A_l \cos(2\pi f_l k / f_s + \phi_l), \quad (5)$$

where $L = p/2 - 1$ is the number of harmonic components, $f_l = l \cdot f_0$ are integer multiples of the fundamental frequency $f_0 = f_s/p = 1/150 \text{ Hz}$ up to the Nyquist frequency $f_N = f_s/2 = 1/2 \text{ Hz}$, A_l are the amplitudes (here uniform, i.e., $A_l = 1/L$), and ϕ_l are the Schroeder phases. The Schroeder phases are computed as

$$\phi_l = \phi_1 - 2\pi \sum_{m=1}^{l-1} (l-m) A_m^2, \quad (6)$$

which produce the multisine signal with the minimum crest factor [3]. A low crest factor maximizes the effective power for a given peak amplitude, and thus maximizes the signal to noise ratio given the input saturation present in the system. The signal is scaled to exploit the full range within the constraints, without being distorted by the saturation.

We choose a periodic multisine for the following reasons:

- **Periodicity** eliminates spectral leakage in the discrete Fourier transform (DFT), because each frequency component falls exactly on a DFT bin.
- **Multiple periods** eliminate leakage when averaging the periods. Averaging is used to reduce the noise variance on the FRF estimate and estimate the noise spectrum. The periodicity can also be exploited to detect transient behavior. Using more periods shortens the period length, reducing frequency resolution. We chose $n = 10$ as a balance between noise variance and frequency resolution.

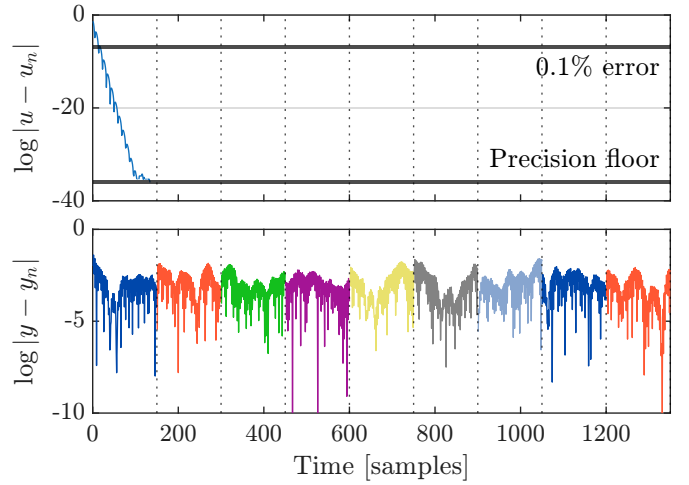


Fig. 2. Transient detection. Log of the difference of each period from the last period. Top the input $u(k)$, bottom the output $y(k)$. The input has a short visible transient, which decays rapidly below 0.1% error.

- **Deterministic amplitude spectrum** provides fully uniform excitation across the frequency band.
- **Schroeder phases** yield a near-optimal crest factor, as opposed to random phases which can produce high peaks.

The Butterworth filter in the system causes the input $u(k)$ exciting $G_0(q)$ to drop off in power above the filter bandwidth. We can flatten the spectrum by increasing the sine magnitude in an inverse relation, but then the overall power of the signal has to be reduced. The inverse of $F(q)$ is unstable, so the multisine also cannot be prefiltered to negate the effect of the filter. As the primary dynamics of G_0 occur below the filter bandwidth, we have chosen to use a flat amplitude spectrum for $r(k)$.

B. Transient Verification

Before estimating the FRF, we verify that transient effects have decayed. We compare each period of $u(k)$ and $y(k)$ against the last period by computing the difference. Fig. 2 shows that there is no apparent transient for $y(k)$, and the input $u(k)$ settles within approximately 20 samples. We therefore use all periods for the FRF estimation, since the transient is short relative to the period length $p = 150$.

C. FRF Estimation

We estimate the FRF using the empirical transfer function estimate (ETFE) method via MATLAB's `etfe()` function. Because our input is exactly periodic with length p over n periods, the FRF is averaged across the periods. The ETFE computes the transfer function directly from the discrete Fourier transform (DFT) as

$$\hat{G}(e^{j\omega_k}) = \frac{\sum_{p=1}^P Y^{[p]}(n)}{\sum_{p=1}^P U^{[p]}(n)}, \quad (7)$$

where $Y^{[p]}(n)$ and $U^{[p]}(n)$ are the Fourier transforms of one period of the output and input signals, respectively.

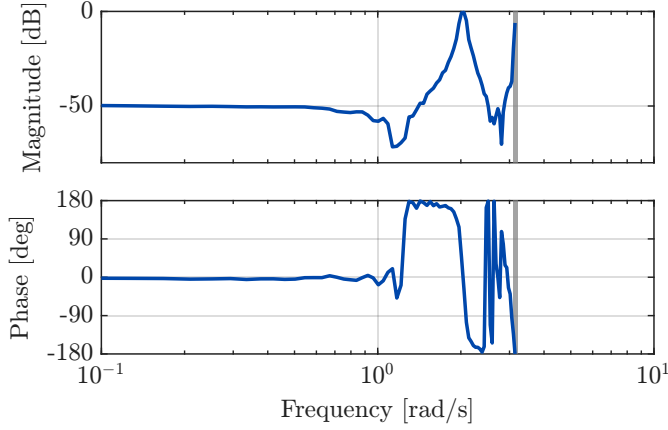


Fig. 3. Bode plot of the ETFE FRF estimate. Visible features include anti-resonances at $\omega \approx 1.2$ and 2.7 rad/s, a resonance at $\omega \approx 2.0$ rad/s, and a DC gain of approximately -50 dB.

We choose ETFE over smoothed spectral analysis (SPA), which computes $\hat{G} = \hat{\Phi}_{yu}(\omega)/\hat{\Phi}_{uu}(\omega)$ using windowed estimates of the cross- and auto-spectral densities. This frequency-domain smoothing can artificially flatten sharp features of the FRF. Because our Schroeder phase multisine excitation is perfectly periodic, the ETFE allows us to achieve noise suppression purely through period averaging, preserving the exact frequency resolution without introducing any windowing bias. Another important advantage of the ETFE used for periodic data is that it is exactly the maximum likelihood estimate of the FRF. Meaning that as the number of periods p increases, the variance of the estimate drops as $\text{Var}(\hat{G}(\omega_k)) \propto 1/P$. As $p \rightarrow \infty$, $\text{Var}(\hat{G}(\omega_k)) \rightarrow 0$ so the estimator is consistent.

The resulting Bode plot in Fig. 3 shows an FRF with a clear structure that is noticeably affected by noise. If the sample size could be increased the FRF estimate would most likely smoothen out. The structure reveals the following features:

- An anti-resonance near $\omega \approx 1.2$ rad/s, indicating a pair of lightly-damped complex zeros,
- A resonance peak near $\omega \approx 2.0$ rad/s, indicating a pair of lightly-damped complex poles,
- A second anti-resonance near $\omega \approx 2.7$ rad/s which is the dominant behavior up to the Nyquist frequency,
- A DC gain of approximately -50 dB.

These features suggest a system with at least one pair of complex poles and two pairs of complex zeros, which will guide the parametric model order selection in Section IV.

D. Noise Spectrum Estimation

We use SPA to estimate the noise power spectrum $\hat{\Phi}_v(\omega)$, which represents the spectral density of the output disturbance $v(k) = H_0(q)e(k)$. For periodic input data, this can be estimated from the residual between the measured output and the FRF-predicted output, averaged over periods. SPA computes it as

$$\hat{\Phi}_v(\omega) = \hat{\Phi}_y(\omega) - \frac{|\hat{\Phi}_{yu}(\omega)|^2}{\hat{\Phi}_u(\omega)} \quad (8)$$

using the estimated auto and cross power spectral densities $\hat{\Phi}(\omega)$ of the output y and input u .

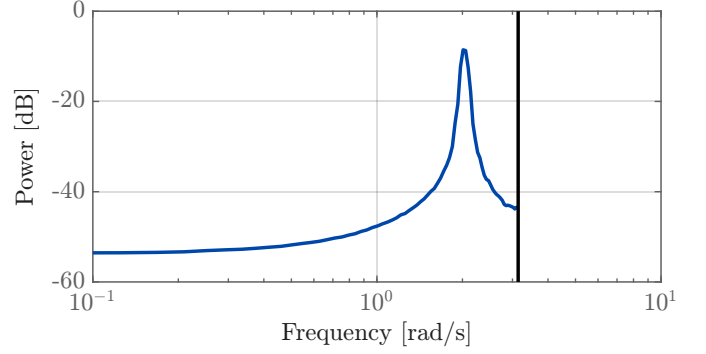


Fig. 4. Estimated noise power spectrum $\hat{\Phi}_v(\omega)$. The noise energy is concentrated around $\omega \approx 2$ rad/s, indicating a colored noise process.

Fig. 4 shows that the noise is *colored*: the energy is concentrated around $\omega \approx 2$ rad/s rather than being uniformly distributed. This means $H_0(q) \neq 1$, and the noise model must be explicitly parametrized to capture this spectral shape. This observation informs the model structure selection in Section IV.

IV. PARAMETRIC IDENTIFICATION AND VALIDATION

In this section, we design a new experiment, select a parametric model structure, estimate the model parameters, and validate the result for consistency and minimum variance.

A. Experiment Design

For parametric identification, we design a new experiment of length $N = 3000$ samples. Since we now do parametric identification, we do no longer need periodicity. Therefore, we do not require a multisine anymore. We now switch to using a pseudo-random binary sequence (PRBS) as reference signal, with amplitude levels $\pm M$. We use PRBS because it is a binary signal, meaning it maximizes the signal power for a given amplitude constraint $|r(k)|$. Next to that, it excites the plant up to the Nyquist frequency, which is necessary for parametric identification where persistence of excitation of sufficiently high order is required.

The collected data of length $N = 3000$ is split into two equal halves: the first $N_e = 1500$ samples for estimation and the remaining $N_v = 1500$ for validation. Both datasets are mean-subtracted using `detrend` to remove any DC offset, which could otherwise bias the identification.

B. Model Structure Selection

We select the Box-Jenkins (BJ) model structure is:

$$y(t) = \sum_{i=1}^{n_u} \frac{B_i(q)}{F_i(q)} u_i(t - n_{ki}) + \frac{C(q)}{D(q)} e(t), \quad (9)$$

where n_u is the number of input channels, $B(q)$ and $F(q)$ parametrize the plant model $G(q) = B(q)/F(q)$, and $C(q)$ and $D(q)$ parametrize the noise model $\hat{H}(q) = C(q)/D(q)$

independently. The orders of the Box-Jenkins model are defined as follows:

$$n_b : B(q) = b_0 + b_1 q^{-1} + \dots + b_{n_b-1} q^{-n_b+1} \quad (10)$$

$$n_c : C(q) = 1 + c_1 q^{-1} + \dots + c_{n_c} q^{-n_c} \quad (11)$$

$$n_d : D(q) = 1 + d_1 q^{-1} + \dots + d_{n_d} q^{-n_d} \quad (12)$$

$$n_f : F(q) = 1 + f_1 q^{-1} + \dots + f_{n_f} q^{-n_f} \quad (13)$$

and n_k denotes the input-output delay in samples. We have selected BJ over others since:

- **ARX** ($y = B/A \cdot u + 1/A \cdot e$): has a shared denominator $A(q)$ for both the plant and noise model which couples them. If the error model order is wrong, the plant estimate will be biased.
- **ARMAX** ($y = B/A \cdot u + C/A \cdot e$): as with the ARX model, the ARMAX model has a shared denominator $A(q)$ for both the plant and noise model which couples them.
- **OE** ($y = B/F \cdot u + e$): assumes white noise at the output ($H_0 = 1$), which contradicts our finding that the noise is colored (Section III).

Compared to these other model structures, BJ separates the plant B/F and noise C/D completely. Even if C/D does not capture the error model, the plant estimate remains consistent as long as the true system $G_0 \in \{B/F\}$.

C. Model Order Selection

Based on the FRF analysis (Section III-C), we select the lower bounds on the model orders $[n_b, n_c, n_d, n_f, n_k]$. Since we have two anti-resonances in the FRF, each corresponding to a pair of complex zeros, we require a $B(q)$ polynomial of at least degree four. As defined in (10), the degree of $B(q)$ is $n_b - 1$, which gives us $n_b \geq 5$. Due to the present resonance peak, we require at least a pair of complex poles in $F(q)$. Due to the low-pass behavior with roll-off at higher frequencies, we require at least two additional poles, resulting in $n_f \geq 4$. Note that to capture roll-off, we also require $n_f > n_b - 1$.

Regarding the error model, the noise power spectrum exhibits strong bandpass characteristics, rolling up from low frequencies to a concentrated peak around 2 rad/s. To model this resonant peak, we need a pair of poles. Therefore, we have $n_d \geq 2$. Furthermore, to capture the low-frequency attenuation and the anti-resonant notches, the noise model must include zeros. Assuming at least one pair of complex zeros is needed to shape this bandpass envelope, we set $n_c \geq 2$.

Finally, the input-output delay n_k represents the dead time of the system. Observing the phase of the plant FRF, the phase angle remains flat near 0 degrees at low frequencies, indicating negligible time delay. Next to this, the estimated impulse response from the data does not deviate from the zero line (except for some noise). Hence, we set $n_k = 0$.

To transition from these initial estimates to a model, we perform a parameter sweep across a grid of polynomial degrees. We evaluate the resulting candidate models on the validation dataset. Our selection criteria prioritize models whose residuals strictly pass the 99% confidence whiteness tests (no significant residual autocorrelations $\hat{R}_{\varepsilon\varepsilon}(\tau)$ outside $\tau = 0$, paired with no significant dependency in $\hat{R}_{\varepsilon u}(\tau)$). Among the models

satisfying these checks, we choose the configuration that maximizes the time-domain validation fit while favoring a minimal model order to prevent over-parameterization. This selection procedure results in our final polynomial orders of $n_b = 5$, $n_c = 2$, $n_d = 5$, $n_f = 5$, and $n_k = 0$.

D. Consistency Validation

When the estimated plant model is consistent, it converges to the true underlying plant as the number of data points approaches infinity. We cannot provide a mathematical proof of consistency since we have a finite dataset. However, we can validate that our data and model satisfy the prerequisites. For this, there are three conditions that must be met. First, the input u must be persistently exciting (PE) of order $\geq n_f + n_b + 1$. Since we use a PRBS input of order N , the input is PE of order $N \gg n_f + n_b + 1$. Second, the true plant must be in the model set ($G_0 \in \mathcal{G}$), and similar for the true noise filter ($H_0 \in \mathcal{H}$), resulting in the full system being captured in the model set ($\mathcal{S} \in \mathcal{M}$). For this to be the case, the model orders must be sufficiently high to capture all plant dynamics. We can practically check this by validating that the prediction error residuals $\varepsilon(k, \hat{G})$ are not correlated with past inputs $u(k)$. We verify this using the sample cross-correlation function

$$\hat{R}_{\varepsilon u}(\tau) = \frac{1}{N_v} \sum_{k=1}^{N_v} \varepsilon(k) u(k - \tau), \quad (14)$$

evaluated on the validation data (not the estimation data, to avoid overfitting). Under the hypothesis that ε and u are uncorrelated, $\hat{R}_{\varepsilon u}(\tau)$ should lie within the 99% confidence bounds $\pm 2.58/\sqrt{N_v}$. Fig. 5 shows that $\hat{R}_{\varepsilon u}(\tau)$ lies within the confidence bounds for all but one lag. This does not strictly prove that no dynamics of $u(t)$ are left unmodelled in the residuals. However, if we would assume a 95% confidence bound, this would be the case. Therefore, we can consider that G_0 is captured by $\mathcal{G}$. Since both criteria are met, the BJ plant estimate $\hat{G}$ is a consistent estimator of the true plant G_0 under the 95% confidence bounds. In case we consider 99% confidence bounds, this would not be the case due to the cross-correlation $\hat{R}_{\varepsilon u}(-36)$ being out of bounds.

We further validate the model through the prediction of the time-domain on the validation data (Fig. 7) and by comparing the parametric Bode plot against the nonparametric ETFE estimate of Section III (Fig. 6).

E. Minimum Variance Analysis

In order for the estimator to lead to a minimum variance estimate, we have to meet a stricter set of conditions compared to proving only consistency. Now we must meet two additional conditions. First, we need the noise to be Gaussian. In order to prove this, we can perform a whiteness test on the residuals $\hat{R}_{\varepsilon\varepsilon}(\tau)$. In the case that the autocorrelation of the residuals

$$\hat{R}_{\varepsilon\varepsilon}(\tau) = \frac{1}{N_v} \sum_{k=1}^{N_v} \varepsilon(k) \varepsilon(k - \tau), \quad (15)$$

are perfectly white (all values except $\hat{R}_{\varepsilon\varepsilon}(0)$ fall within the 99% confidence bounds), the noise model $\hat{H}_0$ has successfully

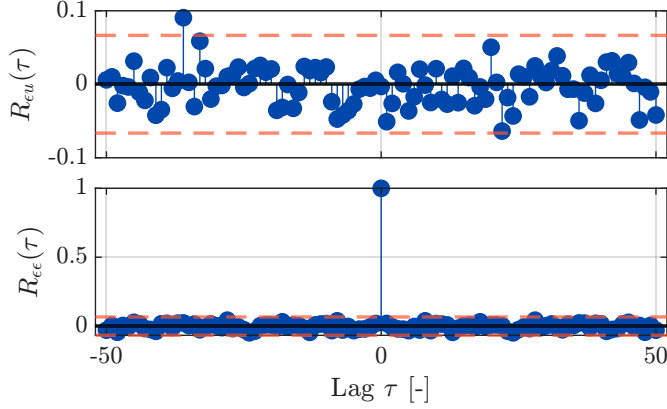


Fig. 5. Residual analysis on validation data. Top: the cross-correlation $\hat{R}_{eu}(\tau)$ lies within the confidence bounds except for $\tau = -36$. Bottom: the residual autocorrelation $\hat{R}_{ee}(\tau)$, showing whiteness of the residuals.

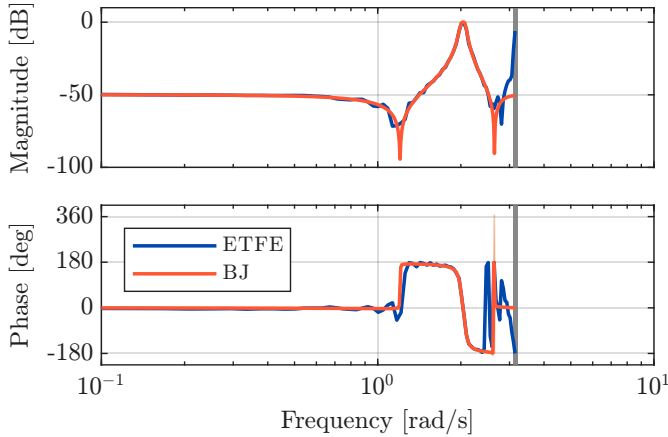


Fig. 6. Bode comparison of the nonparametric ETFE estimate (blue) and the parametric BJ model (red) with 1σ confidence regions. The two estimates agree closely across all frequencies. The confidence region is only visible around the high-frequency anti-resonance.

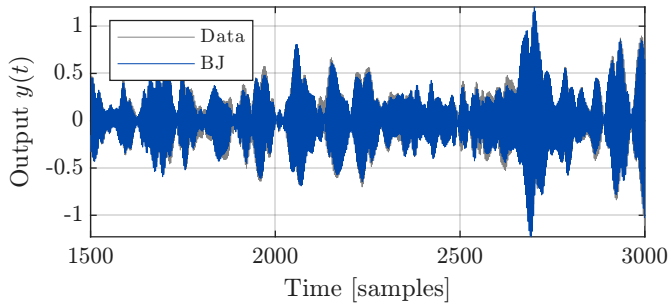


Fig. 7. Time-domain validation: the BJ model output closely tracks the validation data, achieving an 83% fit.

extracted all predictable information from the disturbance. Therefore, the true system $\mathcal{S}$ is captured within our chosen model set $\mathcal{M}$. Second, the estimator must be asymptotically equivalent to the Prediction Error Method (PEM). A maximum likelihood (ML) estimator achieves this. The BJ polynomial model estimator `bj` utilizes an algorithm that asymptotically approximates the ML estimator. Therefore, assuming that the data length N_e is sufficiently large, the Box-Jenkins model yields a consistent and minimum variance estimate

V. EXPERIMENTAL VERIFICATION OF VARIANCE

In this section, we verify the theoretical parameter variance estimates by comparing them against the empirical variance obtained from Monte Carlo simulations.

A. Monte Carlo Simulation Setup

We repeat the parametric identification from Section IV for $n_{MC} = 100$ independent runs. Each run uses the same PRBS reference signal $r(k)$ but generates a new noise realization $e(k)$ through the system function. Using a fixed input across all runs matches the assumption underlying the covariance estimate: the input is deterministic, and only the noise varies. For each run $i = 1, \dots, n_{MC}$, the following operations are performed:

- 1) Generate a new dataset $\{u_i(k), y_i(k)\}$ by passing the fixed $r(k)$ through the system.
- 2) Estimate a BJ model $\hat{G}_i$ with orders $[n_b, n_c, n_d, n_f, n_k]$ on the estimation data.
- 3) Extract the parameter vector θ_i of $\hat{G}_i$ and the diagonal of the covariance matrix via `getcov()`.

B. Parameter Distributions

Fig. 8 shows that the parameters vary across runs. This is expected since each experiment has a different stochastic noise realization $e(k)$, which perturbs the output $y(k)$ and hence the cost function that the PEM optimizer minimizes. The magnitude of the parameter variance depends on the SNR and inversely on the number of estimation samples N_e .

To evaluate the statistical quality of the estimator, the parameter variations are presented as normalized Z-scores. Since BJ utilizes a ML framework, an optimal, unbiased estimator should yield zero-mean, normally distributed parameter errors. The denominator coefficients of $F(q)$, which dictate the system's resonant poles, demonstrate good statistical behavior. Their distributions are symmetrical, centered near zero, and bounded within $\pm 3\sigma$, confirming that the estimator is unbiased when identifying these dynamics. However, the numerator coefficients of $B(q)$, which controls the system zeros and anti-resonances, exhibit systematic bias. Because anti-resonances naturally occur in frequency bands characterized by localized signal attenuation (and thus lower effective SNR), they are more sensitive to specific noise realizations. To minimize the global cost function, the PEM optimizer appears to compensate for these localized, noise-dominated regions, resulting in the offsets observed in the $B(q)$ parameter distributions.

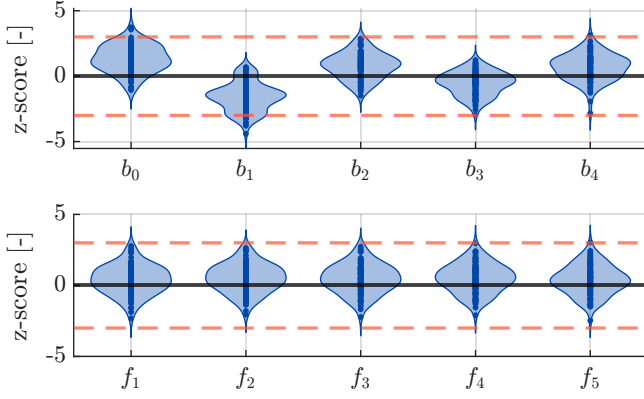


Fig. 8. Standardized Z-scores of the estimated $B(q)$ (top) and $F(q)$ (bottom) coefficients over 100 Monte Carlo runs.

TABLE I
B(Q) AND F(Q) COEFFICIENT MEANS: THEORETICAL VS. MC AVERAGE
OVER 100 RUNS.

Coefficient	Theo. Value	MC Mean	Ratio
b_0	$5.124 \cdot 10^{-3}$	$5.169 \cdot 10^{-3}$	1.009
b_1	$5.371 \cdot 10^{-3}$	$5.298 \cdot 10^{-3}$	0.986
b_2	$3.794 \cdot 10^{-3}$	$3.840 \cdot 10^{-3}$	1.012
b_3	$5.322 \cdot 10^{-3}$	$5.301 \cdot 10^{-3}$	0.996
b_4	$5.122 \cdot 10^{-3}$	$5.166 \cdot 10^{-3}$	1.009
f_1	1.7300	1.7327	1.002
f_2	2.6080	2.6134	1.002
f_3	1.6080	1.6163	1.005
f_4	0.86490	0.87039	1.006
f_5	$-3.079 \cdot 10^{-3}$	$-6.27 \cdot 10^{-4}$	0.204

C. Theoretical Variance

The theoretical covariance matrix P_θ is obtained from a single experiment via `getcov()`, which implements the covariance matrix estimate:

$$P_\theta \approx \frac{\hat{\lambda}}{N_e} \left[\frac{1}{N_e} \sum_{k=1}^{N_e} \psi(k, \hat{\theta}) \psi^\top(k, \hat{\theta}) \right]^{-1}, \quad (16)$$

where $\psi(k, \theta) = -\partial \varepsilon(k, \theta) / \partial \theta$ is the gradient of the prediction error with respect to the parameter vector, and $\hat{\lambda} = (1/N_e) \sum_k \varepsilon^2(k, \hat{\theta})$ is the estimated noise variance. Note that (16) is the estimated Cramér-Rao lower bound (CRLB). For a single experiment, this is only an accurate reflection of the true parameter variance under strict theoretical conditions (see Section IV-E): the estimation data length must be sufficiently large to satisfy asymptotic results, the input must be PE, and the true system must be entirely within the chosen model structure ($\mathcal{S} \in \mathcal{M}$). If unmodeled dynamics persist, and we have structural bias, the residuals $\varepsilon(k, \hat{\theta})$ will not be white. By performing the Monte Carlo simulations, we can bypass these assumptions, providing an empirical benchmark to evaluate whether the theoretical covariance P_θ captures the true uncertainty of the estimator for our finite dataset.

D. Comparison of MC and Theoretical Variance

We compute the variance of each parameter across the 100 runs and compare it against the theoretical variance from a single experiment and the average variance across all runs.

TABLE II
B(Q) AND F(Q) COEFFICIENT VARIANCES: THEORETICAL VS. MC, AND
VARIANCE RATIOS.

Coefficient	Theo. Var.	MC Var.	Ratio
b_0	$1.127 \cdot 10^{-9}$	$1.197 \cdot 10^{-9}$	1.062
b_1	$2.065 \cdot 10^{-9}$	$2.537 \cdot 10^{-9}$	1.229
b_2	$3.327 \cdot 10^{-9}$	$3.149 \cdot 10^{-9}$	0.947
b_3	$1.548 \cdot 10^{-9}$	$1.224 \cdot 10^{-9}$	0.791
b_4	$4.261 \cdot 10^{-9}$	$4.367 \cdot 10^{-9}$	1.025
f_1	$5.569 \cdot 10^{-5}$	$5.669 \cdot 10^{-5}$	1.018
f_2	$1.423 \cdot 10^{-4}$	$1.423 \cdot 10^{-4}$	1.000
f_3	$3.385 \cdot 10^{-4}$	$3.461 \cdot 10^{-4}$	1.022
f_4	$1.233 \cdot 10^{-4}$	$1.252 \cdot 10^{-4}$	1.015
f_5	$4.247 \cdot 10^{-5}$	$4.472 \cdot 10^{-5}$	1.053

Table I and Table II present the complete parameter estimates for the selected BJ model $[5, 2, 5, 5, 0]$, including the theoretical values, the Monte Carlo mean and variance over 100 runs, and the variance ratios. The variance ratio (MC / Theory) quantifies the agreement between the empirical and asymptotic covariance predictions. Both the $B(q)$ and $F(q)$ parameter variances match the theoretical CRLB closely, with ratios near unity. The $F(q)$ ratios are tight (1.000–1.053), while the $B(q)$ ratios show slightly more spread (0.791–1.229) due to the lower effective SNR at the anti-resonance frequencies where the $B(q)$ parameters are most sensitive. In particular, the b_1 and b_3 coefficients have the most spread, indicating the difficulties at the anti-resonance frequencies near $\omega \approx 1.20$ and $\omega \approx 2.63$ rad/s.

This agreement between the MC and theoretical variances validates the asymptotic covariance expression (16) for the current model and dataset size $N_e = 1500$. In particular, the result confirms that the BJ model with orders $[5, 2, 5, 5, 0]$ is not over-parameterized: the linearized CRLB is tight for all parameters, indicating that the first-order Taylor expansion of (16) is a good approximation at this sample size.

VI. MIMO IDENTIFICATION

In this section, we extend the identification workflow to the multiple-input multiple-output (MIMO) case with $n_u = n_y = 2$. In contrast to the SISO case in Section IV, direct polynomial model parametrization becomes less convenient because there are now four transfer functions to estimate from the limited experiment length. We therefore identify the system in state-space form using a two-stage approach: Kung realization (Ho–Kalman adapted to noisy data) to obtain an initial model, followed by PEM state-space refinement (PEM-SS) initialized from that Kung model to obtain a consistent and efficient estimator.

A. MIMO Setup and Experiment Design

The MIMO system is described by

$$y(k) = G_0(q)u(k) + H_0(q)e(k), \quad (17)$$

where $u(k) \in \mathbb{R}^2$, $y(k) \in \mathbb{R}^2$, and $G_0(q)$ is a 2×2 transfer matrix. The experimental setup is shown in Fig. 9. Each reference channel passes through the same Butterworth filter and saturation block as in Section II, after which the two actuator signals excite the MIMO plant.

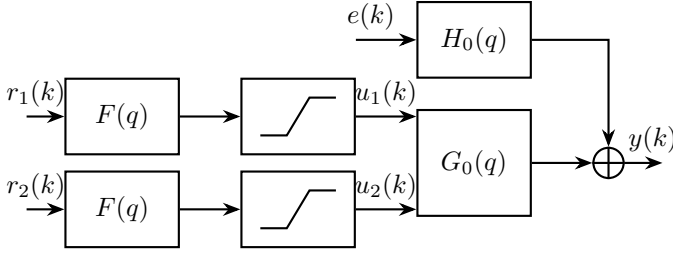


Fig. 9. Block diagram of the MIMO system. The references $r_1(k)$ and $r_2(k)$ pass through the filters $F(q)$ and the saturations before entering the plant $G_0(q)$.

We generate a new reference sequence of length $N = 3000$ with $\text{size}(r) = (N, 2)$. Each input channel is a PRBS signal, and the channels are generated independently. This ensures broad-band excitation per channel while avoiding strong inter-input correlation that would degrade the conditioning of MIMO estimation. The resulting dataset is split into estimation and validation halves ($N_e = 1500$, $N_v = 1500$), and both are detrended before model estimation.

B. Kung Initialization and PEM-SS Refinement

The first stage applies Kung's method in four steps. First, we estimate the first Markov parameters from data by identifying a finite impulse response model using `impulseeest`. Here the estimated impulse-response blocks are

$$g_\tau \in \mathbb{R}^{n_y \times n_u}, \quad \tau = 0, 1, \dots, n_g, \quad (18)$$

with $g_0 = D$.

Second, we build the block Hankel matrix and its one-step shifted counterpart:

$$\mathcal{H}_{n_o, n_c} = \begin{bmatrix} g_1 & g_2 & \cdots & g_{n_c} \\ g_2 & g_3 & \cdots & g_{n_c+1} \\ \vdots & \vdots & \ddots & \vdots \\ g_{n_o} & g_{n_o+1} & \cdots & g_{n_o+n_c-1} \end{bmatrix}, \quad (19)$$

and similarly $\overleftarrow{\mathcal{H}}_{n_o, n_c}$ shifted by one block in time.

Third, we compute the singular value decomposition

$$\mathcal{H}_{n_o, n_c} = U \Sigma V^\top, \quad (20)$$

and choose a reduced rank n from the dominant singular-value drop. Fig. 10 shows a clear drop in significance; we select $n = 6$ in the current implementation.

Fourth, with truncated factors U_n, Σ_n, V_n , we form

$$C = [U_n \Sigma_n^{1/2}]_{1:n_y, :}, \quad B = [\Sigma_n^{1/2} V_n^\top]_{:, 1:n_u}, \quad (21)$$

$$A = \Sigma_n^{-1/2} U_n^\top \overleftarrow{\mathcal{H}}_{n_o, n_c} V_n \Sigma_n^{-1/2}, \quad D = g_0. \quad (22)$$

This yields the initial state-space model (A, B, C, D) from realization.

In the second stage, this realization is used as the initial model for prediction-error minimization in state-space form (`ssest`):

$$\hat{\theta}_{\text{PEM}} = \arg \min_{\theta} \frac{1}{N_e} \sum_{k=1}^{N_e} \varepsilon^\top(k, \theta) \varepsilon(k, \theta), \quad \theta_0 = \hat{\theta}_{\text{Kung}}. \quad (23)$$

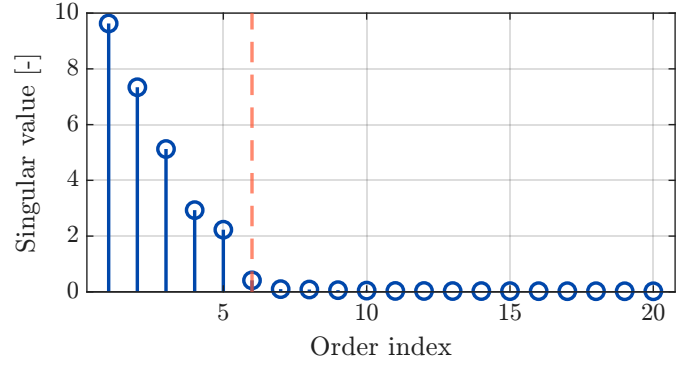


Fig. 10. Singular values of the block Hankel matrix used for Kung realization. The selected rank is based on the dominant singular-value drop.

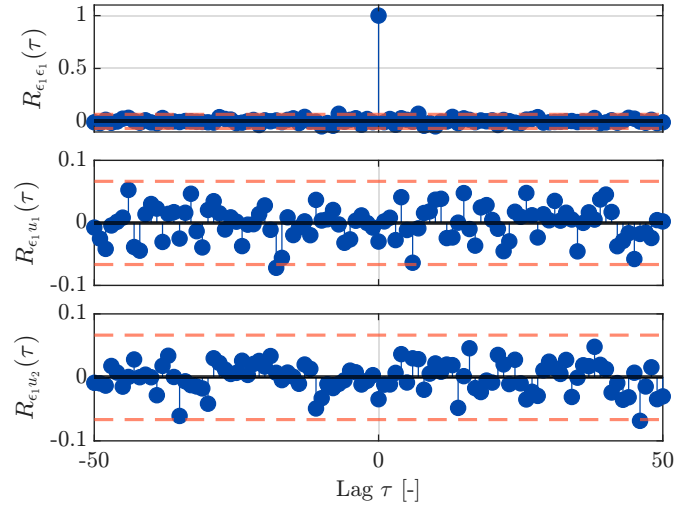


Fig. 11. Residual analysis for y_1 on MIMO validation data. Top: the autocorrelation $\hat{R}_{\varepsilon_1 \varepsilon_1}(\tau)$. Bottom: the residual cross-correlations $\hat{R}_{\varepsilon_1 u_1}(\tau)$ and $\hat{R}_{\varepsilon_1 u_2}(\tau)$. 99% confidence bounds shown as dashed lines.

We refine the estimate from the initial model derived using the Kung algorithm because the initialization provides a reliable non-iterative estimate of the dominant system, but it does not directly optimize the finite-data prediction-error criterion. PEM-SS then refines the state-space structure by minimizing one-step prediction errors on the measured data. This ensures consistency and improves statistical efficiency for finite datasets. Using the Kung model as initialization also provides a physically meaningful starting point for the optimization, which improves numerical robustness and reduces sensitivity to local minima the optimization.

C. Validation and Interpretation

We validate the final PEM-SS model on the independent validation dataset using residual diagnostics and frequency-domain inspection. It was compared to the Kung model, and found to be within 0.01 % of the prediction RMSE, indicating that no real improvement could be made upon the Kung model.

Fig. 11 and Fig. 12 present the residual analysis of the final PEM-SS model. As in the SISO case, the key observation is that the residuals are uncorrelated with past inputs, only spiking to 1 for the autocorrelation at zero lag. Strictly, the

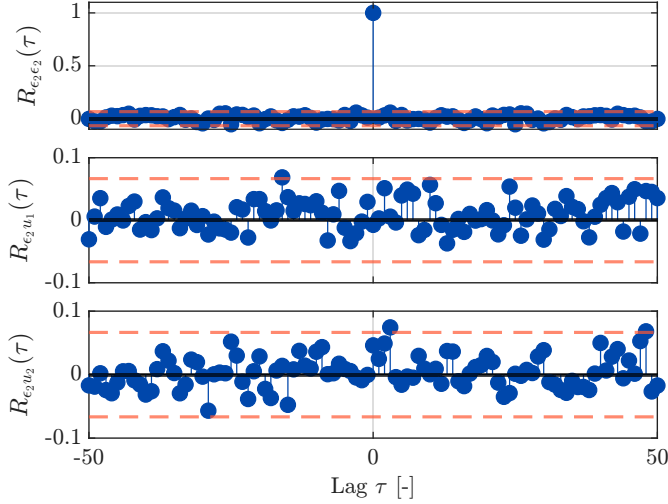


Fig. 12. Residual analysis for y_2 on MIMO validation data. Top: the autocorrelation $\hat{R}_{\epsilon_2 \epsilon_2}(\tau)$. Bottom: the residual cross-correlations $\hat{R}_{\epsilon_2 u_1}(\tau)$ and $\hat{R}_{\epsilon_2 u_2}(\tau)$. 99% confidence bounds shown as dashed lines.

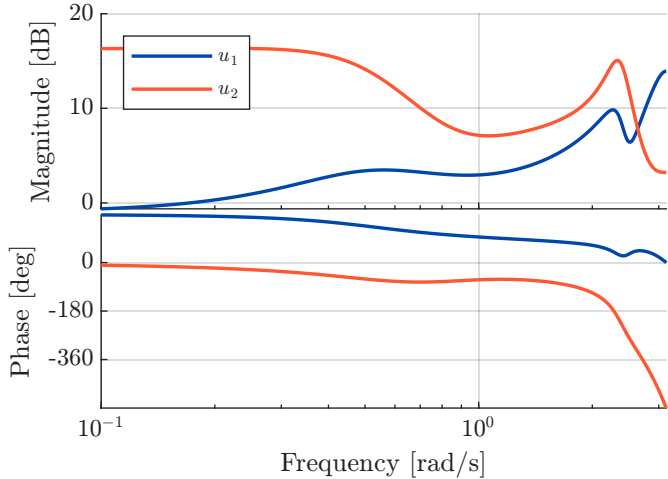


Fig. 13. Bode responses from inputs u_1 and u_2 to output y_1 .

residuals at some lags are just outside the 99% confidence bounds. However, raising the model order does not improve this, so these are not indicative of a model structure error. We thus conclude that the selected state dimension captures the dominant input-output dynamics of the two-channel system, such that the system is in the model class and PEM-SS produces a consistent estimator.

For each output, both input channels are plotted together in a Bode plot. Fig. 13 shows the transfers for output 1, while Fig. 13 shows the transfers for output 2 from both inputs. All transfers exhibit a clear resonance around 2.3 rad s^{-1} , with a more or less clear anti-resonance following and less prominent dynamics present throughout the frequency range.

VII. CONCLUSIONS

In this paper, we systematically applied the system identification cycle to identify accurate data-driven models for both SISO and MIMO systems. First, we characterized the input path, containing a Butterworth low-pass filter and a saturation

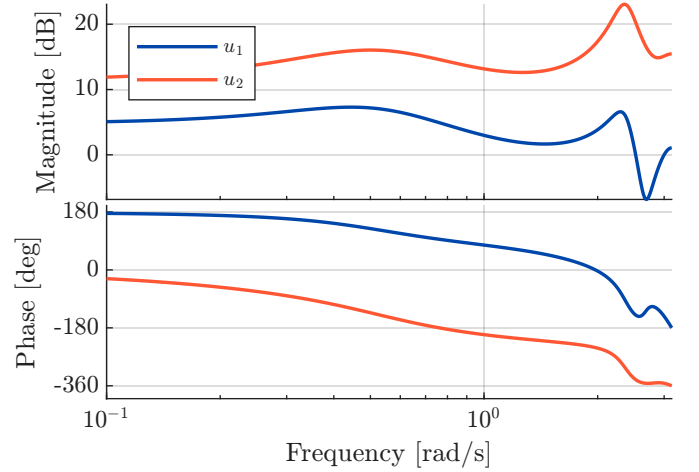


Fig. 14. Bode responses from inputs u_1 and u_2 to output y_2 .

block, which informed the design of excitation signals that maximize input signal-to-noise ratio without inducing distortion. Nonparametric identification using a periodic Schroeder phase multisine provided a quality ETFE frequency response from which the major system dynamics could be identified, even though the relatively small sample-size reduced the absolute quality of the FRF estimate. Estimation of the noise spectrum using SPA showed that the system output is subject to colored noise, motivating the choice of a Box-Jenkins model structure.

For the parametric identification, we estimated a Box-Jenkins model that accurately captures the plant's resonances and anti-resonances. During validation, the residual analysis showed that the autocorrelation was perfectly white. However, we could not strictly prove that the plant estimator is consistent at the 99% confidence level, as one lag of the cross-correlation ($\hat{R}_{\epsilon u}(\tau)$) fell slightly outside the 99% confidence bounds. If the bounds were relaxed to 95%, the cross-correlation would satisfy the threshold, establishing consistency under that condition.

We further validated the theoretical parameter variance of the estimator through Monte Carlo simulations. The empirical variations closely matched the analytical Cramér-Rao lower bounds, confirming that the asymptotic variance expressions hold for our dataset size and that the model is not over-parameterized. Differences in parameter uncertainty could be attributed to local variations in the signal-to-noise ratio, notably around the anti-resonances. Finally, we successfully identified a 2×2 MIMO system, utilizing Kung's realization method from Markov parameters to initialize a state-space prediction error method refinement. Overall, the used experiments and estimation methods yielded accurate representations of the underlying systems.

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